

Logistics Company Data Analysis with SQL

The background features a light pink color with abstract shapes. A large, diagonal, light orange shape runs from the bottom left towards the top right. In the top left, there is a faint, light blue circle. In the bottom right, there is a faint, light orange circle. Stylized, light pink floral or leaf-like motifs are scattered across the background, particularly on the left and right sides.

Uncovering Business Insights
From
Logistics company Data

ABOUT

 **Objective:** The objective of this project is to analyze logistics operations using SQL in order to evaluate workforce distribution, customer segmentation, shipment characteristics, payment behavior, and cost patterns. The analysis supports operational efficiency, delivery performance, and strategic business decisions.

 **Dataset:**

- employee_details** – Employee ID, branch, designation, and name.
- customer** – Customer ID, name, and type.
- shipment_details** - Shipment ID, weight, domain (domestic/international), charges, service type, and content.
- payment_details** – Payment mode, status, amount, and date.
- membership** - Membership start and end dates.
- status** - Shipment status and delivery date.

 **Tools Used:** MySQL Workbench, SQL queries (joins, groupings)

 **Scope of Analysis:**

- Basic: Staffing, Segmentation, Classification ,Frequency
- Intermediate : Averages, Payments, Filtering
- Advanced: Revenue, Behavior , Comparison, Efficiency

Purpose:

- Optimize staff deployment across branches,
- Enhance delivery process efficiency,
- Analyze customer preferences and usage patterns,
- Identify high-value shipments and services,
- Enable informed tactical and strategic decisions through data analysis.

1.Branch-wise Count Of Employees For Efficiency Of Deliveries In Descending Order.

Query:-

```
SELECT E_BRANCH, COUNT(E_ID) AS Employee_Count  
FROM employee_details GROUP BY E_BRANCH ORDER BY Employee_Count DESC;
```

Ans:-

	E_BRANCH	Employee_Count
▶	TX	14
	NY	13
	IL	12
	OH	11
	CA	11
	MA	9

2. CALCULATE THE AVERAGE PAYMENT AMOUNT BASED ON PAYMENT MODE WHERE THE PAYMENT DATE IS NOT NULL.

Query:-

```
SELECT payment_mode, AVG(amount) AS avg_payment  
FROM payment_details WHERE payment_date IS NOT NULL GROUP BY payment_mode;
```

Ans:-

	payment_mode	avg_payment
▶	CARD PAYMENT	45039.8936
	COD	49578.8585

3. RETRIEVE THE NAMES AND DESIGNATIONS OF ALL EMPLOYEES IN THE 'NY' E_BRANCH.

Query:-

```
SELECT E_Name, E_DESIGNATION  
FROM employee_details WHERE E_BRANCH = 'NY';
```

Ans:-

	E_Name	E_DESIGNATION
▶	Bryce	Block development manager
	Ryan	Branch manager
	Earle	Delivery Boy
	Dianne	Branch manager
	Baldemar	Market analyst
	Kim	Office manager

4. CALCULATE THE TOTAL NUMBER OF CUSTOMERS IN EACH C_TYPE (WHOLESALE, RETAIL, INTERNAL GOODS).

Query:-

```
SELECT C_TYPE , COUNT(*) AS total_customers  
FROM customer GROUP BY C_TYPE;
```

Ans:-

	C_TYPE	total_customers
▶	Internal Goods	68
	Wholesale	54
	Retail	78

5. RETRIEVE THE CURRENT STATUS AND DELIVERY DATE OF SHIPMENTS MANAGED BY EMPLOYEES WITH THE DESIGNATION 'DELIVERY BOY'.

Query:-

```
SELECT Status.SH_ID,status.CURRENT_STATUS,status.DELIVERY_DATE  
FROM status JOIN employee_details ON employee_details.E_ID = status.SH_ID  
WHERE E_DESIGNATION = 'Delivery Boy';
```

Ans:-

	SH_ID	CURRENT_STATUS	DELIVERY_DATE
▶	199	NOT DELIVERED	
	515	NOT DELIVERED	
	955	DELIVERED	12/27/1992

6. EXPLORE EMPLOYEE COUNT BASED ON THE DESIGNATION-WISE COUNT OF EMPLOYEES' IDS IN DESCENDING ORDER.

Query:-

```
SELECT E_DESIGNATION, COUNT(E_ID) AS total_employees  
FROM employee_details GROUP BY E_DESIGNATION ORDER BY total_employees DESC;
```

Ans:-

	E_DESIGNATION	total_employees
►	Delivery Boy	14
	Market analyst	11
	Chief finance officer	10
	Transport manager	10
	Warehouse manager	10
	Project director	10

7. CALCULATE THE AVERAGE SHIPMENT WEIGHT BASED ON PAYMENT_STATUS WHERE SHIPMENT CONTENT DOES NOT START WITH "H."

Query:-

```
SELECT Payment_details.PAYMENT_STATUS, AVG(Shipment_details.SH_WEIGHT) AS avg_shipment_weight
FROM shipment_details JOIN payment_details ON payment_details.C_ID = shipment_details.C_ID
WHERE shipment_details.SH_CONTENT NOT LIKE 'H%' GROUP BY payment_details.PAYMENT_STATUS;
```

Ans:-

	PAYMENT_STATUS	avg_shipment_weight
▶	NOT PAID	540.3143
	PAID	515.4533

8. CALCULATE THE AVERAGE SHIPMENT WEIGHT FOR EACH SHIPMENT DOMAIN (INTERNATIONAL AND DOMESTIC).

Query:-

```
SELECT SH_DOMAIN, AVG(SH_WEIGHT) AS avg_weight  
FROM shipment_details GROUP BY SH_DOMAIN;
```

Ans:-

	SH_DOMAIN	avg_weight
▶	Domestic	535.2752
	International	506.1319

9. IDENTIFY THE SHIPMENT WITH THE HIGHEST CHARGES AND THE CORRESPONDING CLIENT'S NAME.

Query:-

```
SELECT customer.C_NAME, shipment_details.SH_ID, SH_CHARGES FROM shipment_details JOIN customer  
ON shipment_details.C_ID = customer.C_ID ORDER BY SH_CHARGES DESC LIMIT 3;
```

Ans:-

	C_NAME	SH_ID	SH_CHARGES
►	Marie	128	1486
	Willie	305	1473
	Taryn	908	1470

10. COUNT THE CUSTOMER BASE BASED ON CUSTOMER TYPE TO IDENTIFY CURRENT CUSTOMER PREFERENCES AND SORT THEM IN DESCENDING ORDER.

Query:-

```
SELECT C_Type, COUNT(*) AS customer_count  
FROM customer GROUP BY C_Type ORDER BY customer_count DESC;
```

Ans:-

	C_Type	customer_count
▶	Retail	78
	Internal Goods	68
	Wholesale	54

11. FIND THE MEMBERSHIP START AND END DATES FOR CUSTOMERS WITH 'PAID' PAYMENT STATUS.

Query:-

```
SELECT Customer.C_ID, membership.START_DATE, membership.END_DATE FROM customer
JOIN membership ON membership.M_ID = customer.C_ID
JOIN payment_details ON Payment_details.C_ID = customer.C_ID
WHERE payment_details.PAYMENT_STATUS = 'Paid';
```

Ans:-

	C_ID	START_DATE	END_DATE
▶	390	2007-09-06	2021-12-25
	249	2001-08-16	2002-05-08
	896	2016-11-01	2021-07-18

12. LIST THE CLIENTS WHO HAVE MADE 'CARD PAYMENT' AND HAVE A 'REGULAR' SERVICE TYPE.

Query:-

```
SELECT DISTINCT customer.C_NAME FROM customer
JOIN payment_details ON customer.C_ID = payment_details.C_ID
JOIN shipment_details ON customer.C_ID = shipment_details.C_ID
WHERE payment_details.Payment_Mode = 'Card Payment'
AND shipment_details.SER_TYPE = 'Regular';
```

Ans:-

	C_NAME
▶	Mitchell
	Italia
	Abby
	Ray
	Tiffany
	Rosalind

13. COUNT THE CUSTOMER BASE BASED ON THEIR STATUS OF PAYMENT IN DESCENDING ORDER.

Query:-

```
SELECT Payment_Status, COUNT(distinct C_ID) AS customer_count FROM payment_details  
GROUP BY Payment_Status ORDER BY customer_count DESC;
```

Ans:-

	Payment_Status	customer_count
▶	NOT PAID	100
	PAID	100

14. COUNT THE CUSTOMER ACCORDING TO SERVICE TYPE IN DESCENDING ORDER OF COUNT.

Query:-

```
SELECT C_TYPE, COUNT(C_ID) AS customer_count  
FROM CUSTOMER GROUP BY C_TYPE ORDER BY customer_count DESC;
```

Ans:-

	C_TYPE	customer_count
▶	Retail	78
	Internal Goods	68
	Wholesale	54

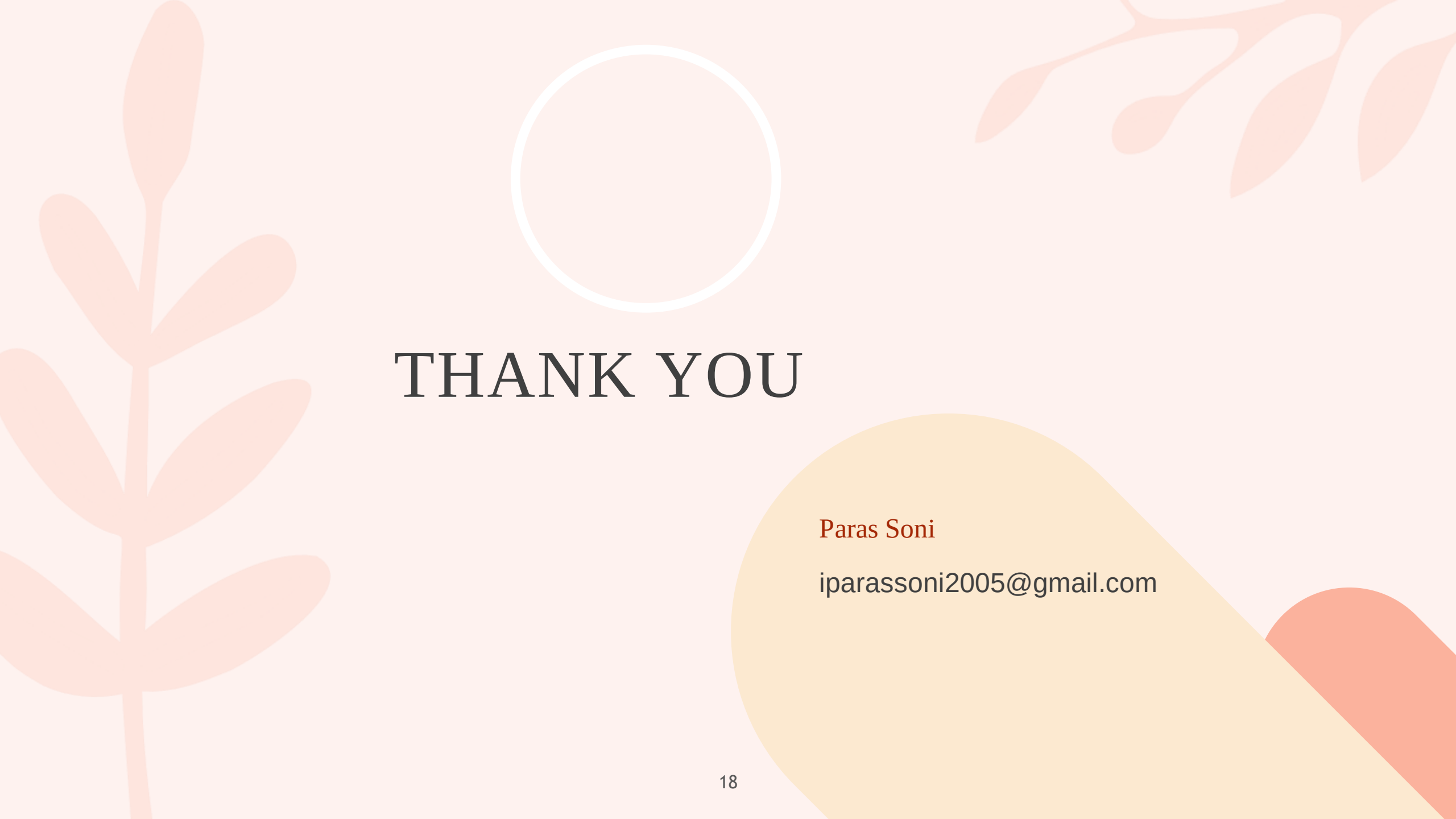
15.COUNT THE CUSTOMERS AS PER SHIPMENT DOMAIN IN DESCENDING ORDER.

Query:-

```
SELECT Shipment_details.SH_DOMAIN, COUNT(DISTINCT Shipment_details.C_ID) AS customer_count  
FROM shipment_details GROUP BY shipment_details.SH_DOMAIN ORDER BY customer_count DESC;
```

Ans:-

	SH_DOMAIN	customer_count
▶	Domestic	109
	International	91



THANK YOU

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